PARKINSON'S DISEASE CLASSIFICATION VIA MULTI-PARAMETRIC ACOUSTIC ANALYSIS AND KERNEL CLASSIFIERS

Author: Aaheri Gangopadhyay
Repository: https://github.com/aaherigangopadhyaystats-code

ABSTRACT

Early detection of Parkinson's Disease (PD) is critical for timely therapeutic intervention and disease management. Vocal impairment(dysphonia) is often one of the earliest signs of neurodegenerative progression. In this project, diagnostic machine learning classifiers were built and evaluated to distinguish individual's with Parkinson's disease from healthy controls using biomedical voice treatments. The project implemented Support Vector Machines (SVM) with Radial Basis Function (RBF) kernels alongside penalized Logistic Regression models. Hyperparameter optimization was conducted via grid search cross-validation across the regularization parameter space. The optimized SVM classifier achieved a diagnostic classification accuracy of 87.2%, demonstrating the effectiveness of vocal acoustic analysis as a non- invasive screening tool.

PROJECT OVERVIEW AND CLINICAL MOTIVATION

Parkinson's disease impairs motor control, leading to progressive changes in speech acoustics such as fundamental frequency variations, vocal tremor and increased noise-to-harmonic ratios. Traditional clinical diagnosis relies on motor assessments, which can be subjective and occur after significant neuronal loss.

This project was structured around three main analytical goals:

- Evaluating biomedical sound measurements to identify key acoustic features strongly correlated with vocal dysphonia.
- Training and fine-tuning non-linear Support Vector Machines (SVM) and linear benchmark classifiers to maximize diagnostic sensitivity.
- Conducting rigorous hyperparameter tuning and sensitivity analysis to balance model accuracy and minimize false negatives in medical screening.

DATASET CHARACTERISTICS AND PREPROCESSING PIPELINE

The biomedical dataset comprised 195 voice recordings from 31 subjects (23 diagnosed with Parkinson's disease and 8 healthy controls). Each record included 22 biomedical voice features:

- MDVP:Fo(Hz), MDVP:Fhi(Hz), MDVP:FIo(Hz): Fundamental acoustic frequency metrics measuring vocal pitch range.

- Jitter metrics (Jitter (%), Jitter (Abs), RAP, PPQ, Jitter: DDP): Measures of variations in fundamental frequency (vocal frequency instability).
- Shimmer metrics (Shimmer, Shimmer(dB), APQ3, APQ5,APQ:DDA): Measures of amplitude variations across consecutive vocal periods (vocal amplitude instability).
- Harmonic-to-Noise Ratios (NHR,HNR): Measures of noise components versus total harmonic structure in the voice signal.
- Non-linear measures (RPDE, D2, DFA, spread1,spread2,PPE): Non-linear dynamical complexity indicators capturing fundamental voice degradation.

PREPROCESSING STEPS

- Feature Scaling: Applied MinMaxScaler to normalize all continuous acoustic attributes into a uniform range [0,1]. This step was vital to prevent features with larger numeric magnitudes (such as raw Hz values) from dominating distance calculations in kernel space.
- Multicollinearity Assessment: Computed by pair-wise feature correlation matrices to identify highly redundant jitter and shimmer sub-metrics.
- Class Balance Inspection: Examined class proportions (-75% PD cases to 25% healthy controls) to ensure performance metrics accounted for class imbalance using balanced accuracy, sensitivity and specificity.

CLASSIFICATION METHODOLOGY AND HYPERPARAMETER OPTIMIZATION

Here two supervised classification approaches to benchmark diagnostic performance were implemented:

Model 1: Penalized Logistic Regression (Linear Baseline)

- Applied L2(Ridge) regularization to prevent overfitting on correlated acoustic features.
- Served as a baseline to evaluate whether linear decision boundaries could sufficiently separate vocal measurements.
- Baseline Accuracy: 82.1%.

Model 2: Support Vector Machine with Radial Basis Function Kernel (SVM-RBF)

- Utilized an RBF kernel
  $K(x, x') = \exp(-\text{gamma} * ||x - x'||^2)$
  project vocal features into a higher-dimensional space where non-linear decision boundaries could separate disease instances.
- Hyperparameter Tuning: Performed systematic Grid Search 5-Fold Cross-Validation over the parameter space:

- Regularization Cost Parameter(c): Evaluated across [0.1,1,10,100]
- Kernel Gamma Parameter: Evaluated across [0.001,0.01,0.1,1, scale].

- Optimal Configuration: Found optimal stability and separation at c=10 and gamma =0.1.

DIAGNOSTIC RESULTS AND SENSITIVITY ANALYSIS

In this project model performance was evaluated on a holdout test set using diagnostic classification metrics:

Model Performance Summary:

- Support Vector Machine (RBF Kernel):
- Overall Classification Accuracy: 87.2%.
- Diagnostic Sensitivity (Recall/ True Positive Rate): 91.3%.
- Diagnostic Specificity (True Negative Rate): 75.0%.
- Area under the ROC curve (ROC-AUC): 0.895.

Key Diagnostic Findings:

- High Sensitivity Analysis: High Sensitivity is essential because in clinical screening, missing a positive case (a false negative) carries a severe penalty. The SVM model achieved a 91.3% sensitivity, correctly identifying the vast majority of individuals exhibiting vocal dysphonia.
- Feature Dominance: Non-linear metrics- specifically Pitch Period Entropy (PPE), Recurrence Period Density (RPDE) and fundamental frequency spread measures- contributed most strongly to separating healthy controls from Parkinson's subjects.
- Linear vs. Non-Linear Decision Boundaries: The RBF SVM outperformed linear logistic regression by 5.1% points in overall accuracy, demonstrating that biomedical voice impairment features interact non-linearly.

CONCLUSIONS

- Non-Invasive Diagnostic Potential: Multi-parametric acoustic analysis of voice recordings provides a strong reliable non-invasive signal for early screening.
- Importance of non-linear kernel space: Non-linear kernel projections effectively map complex interactions between frequency jitter and amplitude shimmer metrics that gets ignored by linear models.

Future extensions: In future, the project aims to explore deep learning architectures and incorporate feature selection algorithms such as Recursive Feature Elimination to construct lightweight, real-time diagnostic voice screening applications.